How the playground models the sea

The air-sea surface, the water body and the measured sea colour, as SMART-G and the playground use them. Written for the project's physics and code, as a reference; November 2026.

In short

- The sea is two things for the light: a **rough surface** that reflects (sun glint, sky light) and refracts, and a **water body** below it that sends part of the entering light back out. The second is what gives the sea its colour.
- The surface is SMART-G's **Cox-Munk rough sea**, set by the wind speed alone, with wave shadowing. It is always there when `surface` is `ocean`.
- The water body has **three settings**: none (the sea only reflects), a **chlorophyll column** (water and particles whose properties follow from one chlorophyll value) or a **measured spectrum** (a diffuse reflector under the surface with a measured remote-sensing reflectance, default: very clear water measured by MOBY).
- The water body works only in the cloud-free scene (`cloud_source: none`): SMART-G's 3D mode cannot hold it.
- On a real MODIS pixel the measured spectrum reaches 0.95 / 0.95 / 0.97 of the observed blue / green / red reflectance, the chlorophyll column 0.90 / 0.93 / 0.98, a reflecting-only sea 0.79 / 0.88 / 0.96.
- Spectra supplied: a very clear-water spectrum (MOBY) and four greener ones by chlorophyll class (NASA NOMAD data). Greener water makes the water's own light green, but the satellite still sees a bluish sea, because the atmosphere supplies most of the blue.
- Known simplifications: the light leaving the water is isotropic (Lambertian) in the measured model; no whitecaps; no wind direction; the conversion constants are approximate.

1 What the sea adds to the satellite signal

Over a cloud-free sea the satellite receives light from three kinds of paths (Figure 1). **A** is sunlight scattered in the atmosphere that never touches the sea; it is the same for any surface. **B** is sunlight reflected by the sea surface: the sun glint near the mirror direction and the reflected sky light everywhere else. **C** is sunlight that crosses the surface, is scattered and absorbed in the water, and comes back out: the *water-leaving light*. Paths B and C both pass back through the atmosphere on the way to the satellite.

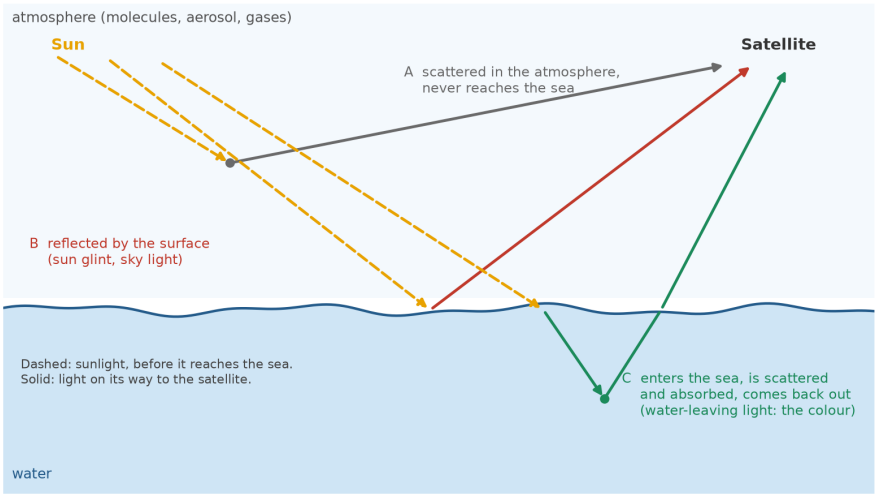


Figure 1. The three kinds of light paths over a cloud-free sea.

Which paths matter depends on the wavelength. Pure water absorbs red light strongly, so path C is almost zero in the red and strongest in the blue. In the MODIS test of section 5 the water-leaving light is about 17% of the blue top-of-atmosphere reflectance, about 7% of the green and about 1% of the red (the difference between the reflecting-only and the measured-spectrum runs). Without a water body only paths A and B exist, and the simulated sea is too dark, most of all in the blue.

2 The air-sea surface

The surface is SMART-G's RoughSurface: a sea whose wave slopes follow the Cox and Munk (1954) statistics, set by the 10 m wind speed (`wind_speed_ms`, default 5 m/s). Light that reaches it is reflected or, when allowed, refracted into the water by the Fresnel laws, with the refractive index of water relative to air (SMART-G's default 1.33). The playground always turns **wave shadowing** on (waves hiding each other at grazing angles).

The published Cox-Munk fit for the total mean-square slope is $\sigma^2 \approx 0.003 + 0.00512\,W$, with W the wind speed in m/s; for W = 4 m/s that is 0.023, an rms slope of about 8.7°, and for 10 m/s 0.054 (13°). I quote the literature fit here and did not check SMART-G's code for it.

	Reflection only (no water body)	**With a water body**
SMART-G setting	`sur=1, brdf=True, wave_shadow=True`	`sur=3, wave_shadow=True`
What happens at the surface	All light is reflected. SMART-G replaces the slope sampling by the Cox-Munk BRDF (its analytic reflectance of the rough sea).	Light is reflected or refracted into the water, photon by photon, on sampled wave slopes; light coming up from the water is refracted out or reflected back down (total internal reflection included).
Below the surface	Nothing: light that would enter is lost, the sea is black.	A water body: the chlorophyll column or the measured reflector (section 3).

Not modelled. Whitecaps and foam (they brighten the sea at high wind, in every band); the wind direction (SMART-G's RoughSurface takes only a speed, so the waves are isotropic); swell and slicks.

3 The water body: three settings

`water_body` switches the water body on (ocean surface and cloud-free scene only). `water_model` then chooses how it is described (Figure 2).

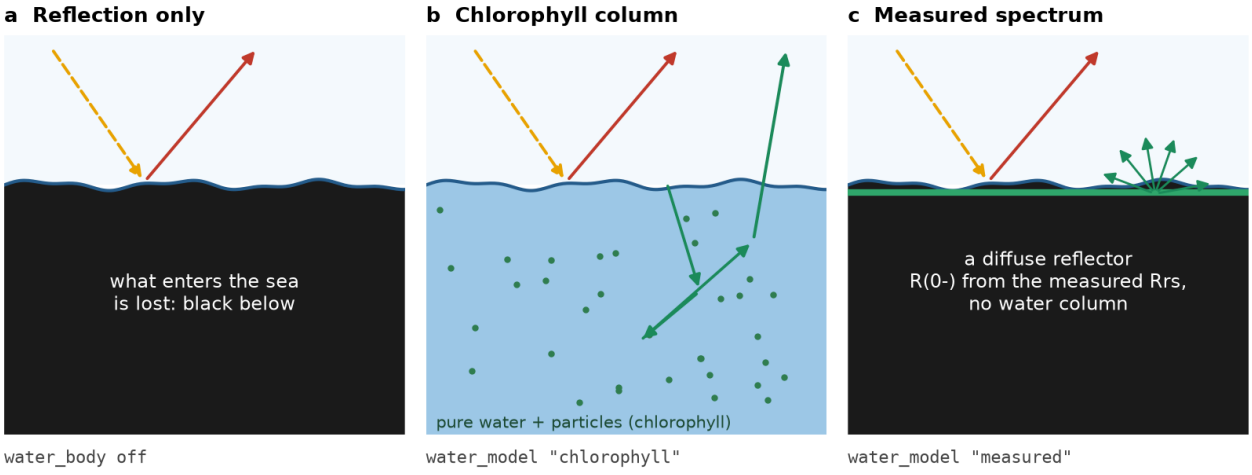


Figure 2. The three ways the sea is described. Dashed: sunlight; red: reflected at the surface; green: water-leaving light.

3.1 Chlorophyll column (`water_model: chlorophyll`)

SMART-G's Water1D with a HydrosolPR, a bio-optical model of open-ocean (Case 1) water in which **one number, the chlorophyll concentration** (`chlorophyll_mg_m3`, default 0.05 mg/m³), sets all the optical properties. The column is uniform with depth, 10 km deep (SMART-G's default grid) with

a black floor, so it is effectively infinitely deep.

Ingredient	How SMART-G describes it
Pure water	Absorption read from SMART-G's auxiliary data; scattering 19.3×10^{-4} $(\lambda/550\ \text{nm})^{-4.3}$ m^{-1}.
Phytoplankton absorption	Bricaud et al. (1998), a function of the chlorophyll concentration.
Coloured dissolved and detrital matter (CDM)	Bricaud et al. (2012).
Particle scattering	$0.416\ \text{chl}^{0.766}$ $(550/\lambda)$ m^{-1} (Loisel and Morel, 1998).
Phase function	Derived from the backscattering ratio as a mixture of two Fournier-Forand functions; the forward peak is truncated (SMART-G's default water truncation).

Strength: the colour follows from physics and the angular distribution of the outgoing light is a result, not an assumption. **Weaknesses:** a single parameter cannot describe water that is not Case 1 (coastal water, sediments, dissolved matter unrelated to chlorophyll); the colour inherits every error of the parameterisation; and it is expensive. At 1e7 photons it took 87 s, against 5 s for a reflecting-only sea; at 5 mg/m³ (coastal water) a run was still unfinished after 46 minutes and was stopped.

3.2 Measured spectrum (`water_model: measured`)

Instead of computing the colour, the model is **given** it: the remote-sensing reflectance Rrs(λ) of the sea, measured. SMART-G's `WaterRw` represents the water as a single diffuse (Lambertian) reflector just under the surface, with no water column and no thickness. The surface above it still does the refraction and Fresnel reflection (`sur`=3), so the transmission through the surface is computed, not assumed.

What quantity to give it. `WaterRw` takes the subsurface irradiance reflectance $R(0^-) = Eu(0^-)/Ed(0^-)$, measured just *below* the surface. Published spectra are Rrs, radiance over irradiance just *above* the surface. SMART-G's documentation warns that giving it Rrs or the water-leaving reflectance would count the transmission through the surface twice. The playground therefore converts, in two steps (`subsurface_reflectance()` in `playground_core.py`):

```
(1) subsurface rrs = Rrs / (ζ + Γ Rrs),   the inverse of Rrs = ζ rrs / (1 − Γ rrs), with ζ = 0.52
and Γ = 1.7 (the relation of Lee et al., 2002);
(2) R(0⁻) = Q × rrs,   with Q = Eu/Lu = 3.4.
```

The constants ζ, Γ and Q were written from memory of the literature and were not checked against the papers. They are approximate; I estimate a few per cent in $R(0^-)$, no more. They are named constants at the top of `playground_core.py`.

3.3 The Lambertian approximation, and what the Q factor does

Real water sends slightly more light straight up than sideways; the ratio of the upwelling irradiance to the upwelling radiance in one direction is Q = Eu/Lu, near 3.4 for clear water and varying with the sun and view angles. A Lambertian reflector is isotropic and has $Q = \pi \approx 3.14$. In the measured model the light leaving the water is therefore **isotropic, whatever the geometry**: its angular shape is an approximation. The interface above it is not affected: the glint and the Fresnel reflection stay anisotropic and polarizing.

Q is one number that scales the whole spectrum, so it changes the brightness of the water-leaving light, not the colour. Matching the radiance toward a near-nadir satellite would need $Q = \pi$ instead of 3.4, which would lower the water-leaving light by about 8%. In the MODIS test that is about 1.4% of the blue top-of-atmosphere reflectance, less in green, almost nothing in red; it is much smaller than the remaining 5% gap to the observations, and the constant was left at 3.4. A sun- and view-dependent Q from published tables would be a refinement, not a change of kind. A model with a true angular distribution is the chlorophyll column, at the cost of a bio-optical model and run time.

4 The spectra supplied

4.1 Very clear water: MOBY

The default file, `configs/sea_colour/moby_clear_water_rrs.csv`, is the mean of two spectra measured by the Marine Optical Buoy (MOBY) west of Lanai, Hawaii (20.49°N, 157.12°W), on 2002-07-15 and 2009-05-18, every 5 nm from 380 to 730 nm. MOBY sits in very clear, low-chlorophyll water. The spectra come from NASA's SeaBASS archive (hyperspectral Rrs examples file, columns 7 and 8); the file header gives the credit. Outside 380-730 nm the model holds the end values constant.

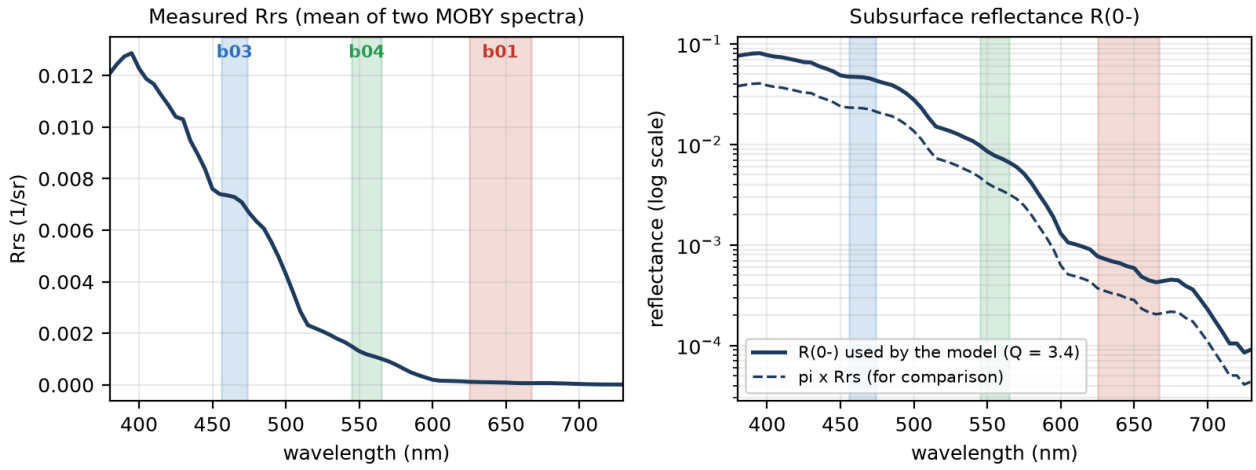


Figure 3. Left: the measured Rrs, with the MODIS bands used in the test. Right: the subsurface reflectance R(0⁻) that the model derives (log scale); the dashed curve is π × Rrs.

λ (nm)	410	440	470	490	550	670
Rrs (1/sr)	0.01168	0.00895	0.00709	0.00556	0.00131	0.00007
R(0⁻) used	0.0735	0.0569	0.0453	0.0357	0.0085	0.0004

The shape is what gives the colour: a lot of blue, about a seventh of it in the green, almost nothing in the red. A different water type (greener, with more chlorophyll or sediment) needs its own spectrum; the file format is two columns, wavelength in nm and Rrs in 1/sr, with # comment lines.

4.2 Greener water: spectra by chlorophyll class (NOMAD)

For water with more chlorophyll the playground ships four spectra built from **NOMAD**, the NASA bio-Optical Marine Algorithm Dataset (version 2.0 alpha, 18 July 2008): in-situ stations with the water-leaving radiance, the downwelling irradiance and a measured chlorophyll a (fluorometric, HPLC where that is missing). Rrs is the ratio of the first two. Of 4,459 records, 1,016 have all six bands (411, 443, 489, 510, 555 and 670 nm) and a chlorophyll value. They were grouped by chlorophyll and the **band-by-band median** of each group was written to `configs/sea_colour/nomad_chl_<class>.csv` (Figure 4, Table).

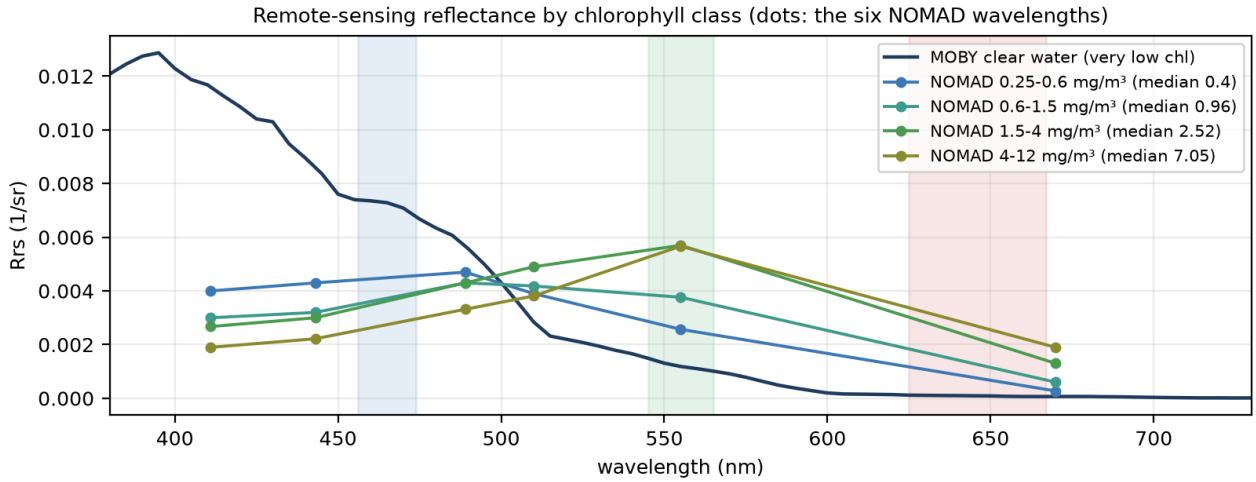


Figure 4. The five spectra supplied. The NOMAD spectra have six wavelengths; the model interpolates linearly between them and holds the end values outside 411-670 nm.

Chlorophyll class (mg/m³)	Stations	Median chl	Rrs 443	Rrs 555	Rrs 443 / Rrs 555
0.25 - 0.6	140	0.40	0.0043	0.0026	1.67
0.6 - 1.5	225	0.96	0.0032	0.0038	0.85
1.5 - 4	207	2.52	0.0030	0.0057	0.53
4 - 12	156	7.05	0.0022	0.0057	0.39
(MOBY, very clear water)	2 spectra	very low	0.0086	0.0012	7.2

What these are not. Each file is a median taken band by band over many stations, a representative shape and not one station's spectrum. The spread is large: in the 1.5-4 class Rrs 555 runs from 0.0027 to 0.0097 between the quartiles, and the higher classes mix in coastal water whose colour is not set by chlorophyll alone. NOMAD 2.0 is an alpha (pre-release) version, and its header limits it to research use. The clearest class (0.02-0.06 mg/m³) has only 17 stations and was not used; MOBY covers that case.

What they do to the colour. The simulation was run for the MODIS test geometry with each spectrum (Figure 5). The water's own contribution, obtained by subtracting the reflecting-only run, turns from blue to green as the chlorophyll rises: the green-to-blue ratio of the light the water adds goes from 0.17 (MOBY) to 0.63, 1.08, 1.65 and 2.2 for the four classes. The satellite, however, still sees a bluish sea: the atmosphere and the surface reflection supply about 0.097 of blue and 0.048 of green whatever the water does, so the green-to-blue ratio of the top-of-atmosphere reflectance only moves from 0.43 to 0.61. Real satellite images of green water look green after a contrast stretch or an atmospheric correction, which the display here does not apply.

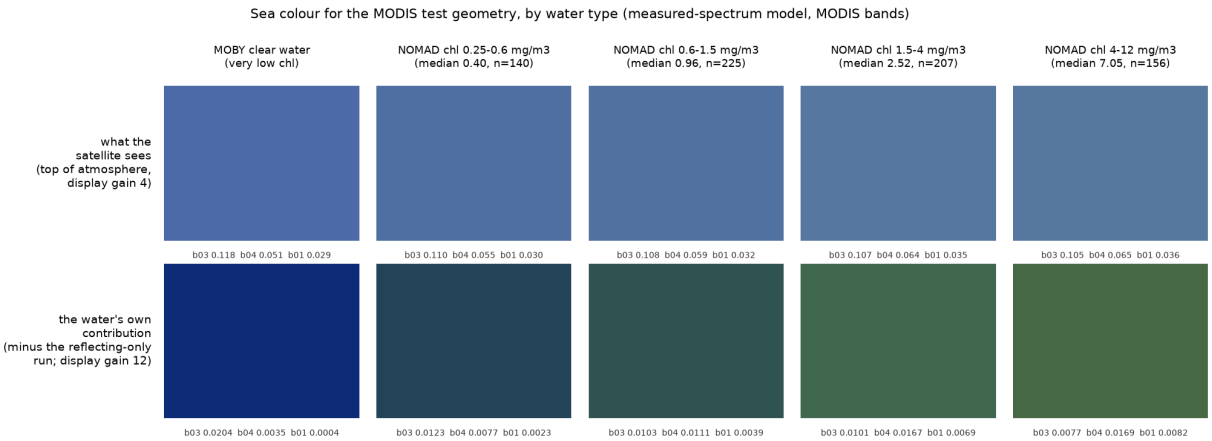


Figure 5. Top: the sea as the satellite sees it for each spectrum (display gain 4; numbers are the top-of-atmosphere reflectance in MODIS bands b03, b04, b01). Bottom: the water's own contribution, the difference between the run and the reflecting-only run (display gain 12). Each run has 1e7 photons, so the differences in the bottom row carry a noise of about ±0.001, large for the small numbers (the green of the MOBY case, 0.0035, is uncertain by about a third).

These runs have no observation to compare with, so they show what the spectra imply, not that the colours are right.

5 Does it give the right colour? A test on a real MODIS pixel

The test reproduces one MODIS Terra pixel of a cloud-free ocean scene in the South Pacific gyre (2023-03-05, 17:50 UTC; latitude −25.2°): sun zenith 39.2°, view zenith 12.9°, wind 4 m/s (a placeholder, not measured), a `maritime_clean` aerosol of optical depth 0.087 (the aerosol type is a guess), ozone 263 DU, water vapour 3.05 g/cm² and pressure 1019 hPa (from MODIS ancillary products). The observed reflectance is the median of 95,463 cloud-screened, glint-free pixels. Simulations use 1e7 photons (about 1% noise).

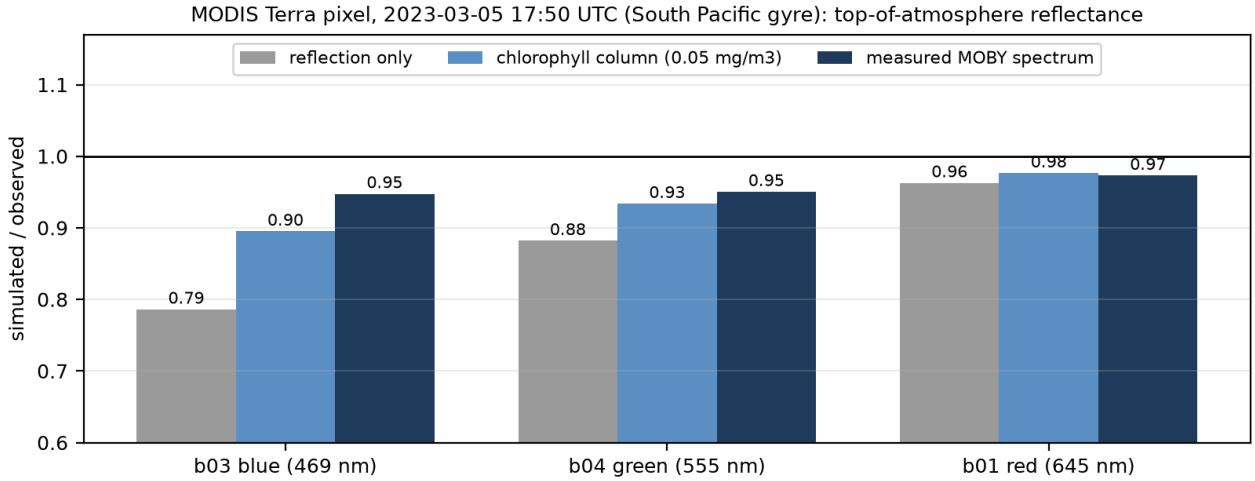


Figure 6. Simulated over observed top-of-atmosphere reflectance in the three MODIS bands, for the three ways of treating the water body.

	b03 blue	b04 green	b01 red	Time (1e7 photons)
Observed (MODIS)	0.1239	0.0541	0.0292	
Reflection only	0.0974 (0.79)	0.0477 (0.88)	0.0281 (0.96)	5 s
Chlorophyll column, 0.05 mg/m³	0.1109 (0.90)	0.0505 (0.93)	0.0285 (0.98)	87 s

	b03 blue	b04 green	b01 red	Time (1e7 photons)
Measured MOBY spectrum	0.1174 (0.95)	0.0514 (0.95)	0.0284 (0.97)	33 s*

Simulated reflectance, and the ratio to the observed value in brackets. *The first run of a session includes start-up; in a later test at 2e7 photons the measured model took 6 s and the chlorophyll column 123 s.

Reading it. The water body is needed: without it the sea is 21% too dark in blue. The measured spectrum is closer than the chlorophyll column and about an order of magnitude faster. A gap of about 5% remains in blue and green. It has not been explained; the candidates are whitecaps (not modelled), the aerosol type and optical depth, the wind speed, the cloud screening of the observation, and the fact that the MOBY spectrum was measured in Hawaii, not at this pixel. The red channel agrees to 3% in all three cases because the water adds almost nothing there. I have not checked whether MODIS reflectance, as stored, is divided by the cosine of the sun zenith, a point that affects every ratio above.

6 The water body and polarization

The polarization of the signal is computed whatever the model. I compared the three settings on the same pixel at 2e7 photons, with the Stokes parameters saved. The polarized signal $\sqrt{(Q^2+U^2)}$ changes by only 1-3% with the water body: the light from the water adds almost no Q or U, so it acts as unpolarized light. It dilutes the degree of linear polarization (DoLP) because it adds to the total intensity I.

	Blue I	Blue DoLP	Blue polarized signal	Green DoLP	Red DoLP	AoLP (all bands)
Reflection only	0.0972	13.0%	0.01259	14.6%	15.9%	−47° to −49°
Measured spectrum	0.1177	10.8%	0.01268	13.6%	15.7%	−47° to −49°
Chlorophyll column	0.1098	11.8%	0.01300	14.1%	15.8%	−47° to −48°

So for polarized images the water model matters through its **intensity**: the one that gets I right (here the measured spectrum) gets the DoLP right. The angle of polarization is unchanged. The limits: one geometry (sun zenith 39°, view 13° from nadir), one noise realization, and no estimate of the Q/U noise at this photon count, so the 2-3% differences in the polarized signal may be noise. The water's own polarization could matter more near the glint or at large view angles; I have not tested that. The AoLP is measured from the meridian plane of the viewing direction.

7 Limits and open points

- **Cloud-free scenes only.** SMART-G's 3D mode fails with a water body (`KeyError: 'iabs_oc'`, in the installed version and in 2.0.6). Cloud scenes have the reflecting sea only. Whether the 3D mode accepts `WaterRw` has not been tested.
- **Isotropic water-leaving light** in the measured model (section 3.3); the angular shape is right only in the chlorophyll column.
- **No whitecaps, no wind direction, no sea floor.** The chlorophyll column is uniform in depth and effectively infinite; the measured model has no depth at all.
- **The conversion constants** (ζ, Γ, Q) are from memory and approximate. The result is a spectrum-shaped, correctly scaled water-leaving radiance to a few per cent, not a validated one.
- **Few spectra.** One very clear-water spectrum (MOBY, two measurements) and four medians by chlorophyll class (NOMAD, six wavelengths each). Other water types need their own data: in-situ archives (SeaBASS, AERONET-OC, BOUSSOLE), satellite Rrs, or simulated sets. A spectrum taken from a satellite ocean-colour product has already been through that product's atmospheric correction. The Lambertian water-leaving light and the constants of section 3.2 were set for clear water and may be less accurate in the greener, murkier classes.

- **The unexplained 5% in blue and green** of the MODIS test (section 5).

8 Using it

In the GUI: set **Cloud** to *none*, **Surface** to *ocean*, tick **Sea water body**, then choose the **Water model** and, for the measured one, the spectrum file (the chlorophyll field is used by the chlorophyll column only). In a scenario file for `run_from_config.py`:

Setting	Meaning
`cloud_source: "none"`, `surface: "ocean"`	The cloud-free ocean scene; the surface needs the wind speed `wind_speed_ms`.
`water_body: true`	Light enters the sea. False: reflection only.
`water_model`	`"chlorophyll"` (default) or `"measured"`.
`chlorophyll_mg_m3`	Chlorophyll of the column, 0.001 to 10 (default 0.05); used by the chlorophyll model only.
`sea_spectrum_file`	CSV of wavelength (nm) and Rrs (1/sr); default `configs/sea_colour/moby_clear_water_rrs.csv`; the greener ones are `configs/sea_colour/nomad_chl_*.csv`; a relative path starts in the program's folder.

A ready example for the MODIS pixel of section 5 is `configs/modis_pixel_measured_sea.json`. A run takes tens of seconds with the measured spectrum at 1e7 photons; about 1e8 photons (400,000 per pixel on 16×16 pixels) gives about 1% noise or better. The images of a cloud-free scene are uniform patches of colour; the numbers to read are in the saved `.npy` files.

References and sources

Cox, C. and Munk, W. (1954), Measurement of the roughness of the sea surface from photographs of the Sun's glitter, J. Opt. Soc. Am. 44, 838-850.

Lee, Z.-P., Carder, K. L., Mobley, C. D., Steward, R. G. and Patch, J. S. (2002), Deriving inherent optical properties from water color: a multiband quasi-analytical algorithm, Appl. Opt. 41, 5755 (the Rrs / rrs relation; constants quoted from memory).

Bricaud, A., Morel, A., Babin, M., Allali, K. and Claustre, H. (1998), Variations of light absorption by suspended particles with chlorophyll a concentration in oceanic (case 1) waters, J. Geophys. Res. 103, 31033-31044.

Bricaud, A., Ciotti, A. M. and Gentili, B. (2012), Spatial-temporal variations in phytoplankton size and colored detrital matter absorption at global and regional scales, Global Biogeochem. Cycles 26, GB1010.

Loisel, H. and Morel, A. (1998), Light scattering and chlorophyll concentration in case 1 waters: a reexamination, Limnol. Oceanogr. 43, 847-858 (as cited in SMART-G's `HydrosolPR`).

The citations of Cox and Munk, Lee et al. and Loisel and Morel (titles, journals, pages) were written from memory and not checked; those of Bricaud et al. are as given in SMART-G's own documentation.

SMART-G documentation and code (`smartg/surface.py`, `smartg/water.py`, v2.0.6): `RoughSurface`, `Water1D`, `HydrosolPR`, `WaterRw`.

Werdell, P. J. and Bailey, S. W. (2005), An improved in situ bio-optical data set for ocean color algorithm development and satellite data product validation, Remote Sens. Environ. 98(1), 122-140 (the citation requested in the NOMAD file). NOMAD v2.0 alpha: https://seabass.gsfc.nasa.gov/wiki/NOMAD/nomad_seabass_v2.a_2008200.txt; the data contributors are named in that file.

MOBY spectra: NOAA MOBY Operations Team, through NASA SeaBASS, https://seabass.gsfc.nasa.gov/wiki/Hyperspectral_Rrs_Examples (file hyperspectral_examples_v1.0.csv, selection and interpolation by C. W. Proctor).

Results of sections 5 and 6: simulations with `playground_core.py` at the commit of this document (Nov 2026); MODIS observation from configs/clear_ocean_MOD021KM_A2023064.1750.json (see CLEAR_OCEAN_COMPARISON.md).